RNA-seq Training: Differential Gene Expression in *E. coli*

Wild type (ECWT) vs *slyA* mutant (ECMS)

A hands-on bioinformatics workshop that takes you through a complete RNA-seq analysis: from downloading raw sequencing reads to finding the genes that change between two bacterial strains. *Escherichia coli* has a small genome (about 4.6 Mb), so every step runs on an ordinary laptop.

Contents

- The data
- What you will learn
- Repository structure
- Setup
- Quick start
- Step 1 – Download the reads
- Step 2 – Reference genome
- Step 3 – Quality control
- Step 4 – Trimming
- Step 5 – Indexing, alignment and gene counting
- Step 6 – Merge the gene counts
- Step 7 – Differential expression
- Questions
- Troubleshooting
- Data source

The data

The data are from NCBI GEO GSE168963 (ENA project PRJNA714667). There are 16 RNA-seq samples of *E. coli* BW25113 (runs `SRR13970433`–`SRR13970448`), each sequenced paired-end (2 FASTQ files per sample, 32 files):

Group	Strain	Samples
ECWT	Wild type	8
ECMS	*slyA* mutant	8

TABLE 1 Summary of *E. coli* BW25113 wild-type and *slyA* mutant strain transcriptome sequencing sample data

Sample description[a]	Corresponding SRA description	No. of reads	% mapped reads[b]	SRA accession no.	GEO accession no.[c]
ECWT1.1_rep1	ECWT1_1	2,888,920	79	SRX10348166	GSM5172890
ECWT1.1_rep2	ECWT1_2	2,864,261	79	SRX10348167	GSM5172891
ECWT1.1_rep3	ECWT1_3	2,803,942	79	SRX10348168	GSM5172892
ECWT1.1_rep4	ECWT1_4	2,838,406	79	SRX10348169	GSM5172893
ECWT1.2_rep1	ECWT1_5	3,097,616	79	SRX10348170	GSM5172894
ECWT1.2_rep2	ECWT1_6	3,065,011	79	SRX10348171	GSM5172895
ECWT1.2_rep3	ECWT1_7	3,000,499	79	SRX10348172	GSM5172896
ECWT1.2_rep4	ECWT1_8	3,026,276	79	SRX10348173	GSM5172897
ECMS2.1_rep1	ECMS2_1	2,543,159	82	SRX10348174	GSM5172898
ECMS2.1_rep2	ECMS2_2	2,507,462	82	SRX10348175	GSM5172899
ECMS2.1_rep3	ECMS2_3	2,419,332	82	SRX10348176	GSM5172900
ECMS2.1_rep4	ECMS2_4	2,438,239	82	SRX10348177	GSM5172901
ECMS2.2_rep1	ECMS2_5	3,501,908	83	SRX10348178	GSM5172902
ECMS2.2_rep2	ECMS2_6	3,464,396	84	SRX10348179	GSM5172903
ECMS2.2_rep3	ECMS2_7	3,398,574	83	SRX10348180	GSM5172904
ECMS2.2_rep4	ECMS2_8	3,417,567	85	SRX10348181	GSM5172905

[a] *E. coli* BW25113 wild-type samples are denoted ECWT1, and its *slyA* mutant samples are denoted ECMS2. The total number of reads for each of the samples are as follows: ECWT1.1, 11,395,529; ECWT1.2, 12,189,402; ECMS2.1, 9,908,192; and ECMS2.2, 13,782,445.

[b] Percentage of reads mapped to the reference genome (*E. coli* BW25113; GenBank accession number CP009273.1).

[c] The GEO accession number of the transcriptome series is GSE168963, and the SRA BioProject number is PRJNA714667. The GEO title of the project is "Transcriptome RNA Sequencing Data Set of Differential Gene Expression in *Escherichia coli* BW25113 SlyA Wild Type and Mutant Strains."

Table 1 from the dataset's publication: the 16 samples, their read counts and mapping rates.

What you will learn

Step	Task	Tool	Script	Output
1	Download reads	ENA, `wget`	`download_fastq.sh`	32 FASTQ files
2	Reference genome	NCBI	`get_reference.sh`	Genome FASTA + GTF
3	Quality control	FastQC, MultiQC	`fast_qc.sh`	Quality reports
4	Trimming	Trim Galore	`trim.sh`	Clean FASTQ files
5	Indexing, alignment, gene counting	STAR	`ecoli_gene_quantification.sh`	BAM files + gene counts
6	Merge gene counts	R	`merge_data.R`	One count table

Step	Task	Tool	Script	Output
7	Differential expression	R, DESeq2	`DEG_ecoli.R`	Results table, volcano plot, heatmap, PCA

Steps 1–5 run in a Linux terminal (Ubuntu, or WSL on Windows). Steps 6–7 run in R or RStudio.

Repository structure

```
training/
├── README.md              # this tutorial
├── scripts/               # one script per step
├── reference/             # E. coli genome (FASTA) and annotation (GTF)
├── DEG/                   # pdata.csv (sample groups) + results of Steps 6-7
├── Document/              # practical document, paper, images
└── raw_data/              # reads and Step 3-5 results (created locally, NOT
on GitHub)
    ├── fastqc_results/
    ├── trimmed_reads/
    ├── star_index/
    └── star_align/
```

`raw_data/` is excluded from GitHub (see `.gitignore`) because the reads and alignments are several GB. Only an empty placeholder (`raw_data/.gitkeep`) is kept, so the folder exists after cloning. The scripts recreate everything in it.

Run every command from the `training/` folder. Each script works out where `training/` is from its own location, so the folder can be anywhere on your computer.

Setup

1. Get the material

```
git clone https://github.com/<your-username>/<repo-name>.git training
cd training
mkdir -p raw_data   # only needed if the folder is missing
```

2. Install the command-line tools

```
sudo apt update
sudo apt install fastqc cutadapt trim-galore rna-star multiqc
```

If `multiqc` is not available through `apt`, install it with `pip install multiqc`. Check that the tools work:

```
fastqc --version && cutadapt --version && trim_galore --version && STAR --
version && multiqc --version
```

3. Install the R packages

In R or RStudio:

```
install.packages(c("tidyverse", "BiocManager", "pheatmap"))
BiocManager::install(c("rtracklayer", "DESeq2", "EnhancedVolcano"))
```

4. Set your folder in the R scripts

The two R scripts start with this line. Change it to wherever your `training/` folder is:

```
training_dir <- path.expand("~/Desktop/bioinformatics_workshop/training")
```

5. Match the number of CPU cores (optional)

The scripts use 4 threads for FastQC and Trim Galore and 8 for STAR (`THREADS=8` at the top of `ecoli_gene_quantification.sh`). Check how many cores you have with `nproc`, and lower these numbers if you have fewer.

Quick start

To run the whole analysis in one go:

```
bash scripts/download_fastq.sh
bash scripts/get_reference.sh
bash scripts/fast_qc.sh
bash scripts/trim.sh
bash scripts/ecoli_gene_quantification.sh
Rscript scripts/merge_data.R
Rscript scripts/DEG_ecoli.R
```

In the workshop, run the steps one at a time and look at the output of each one before moving on.

> **Tip:** The download and alignment steps take a while. Run them inside `screen` or `tmux`, or with `nohup`, so they keep going if the terminal closes: `nohup bash scripts/download_fastq.sh > download.log 2>&1 &`

Step 1 – Download the reads

```
bash scripts/download_fastq.sh
```

The script downloads the 32 FASTQ files from ENA into `raw_data/`. It:

- **resumes** files that were only partly downloaded (`wget -c`), so it is safe to run again;
- skips files that are already complete;
- **checks** every file with `gzip -t` at the end.

The last lines of the output tell you whether everything worked:

```
FASTQ files present: 32 of 32
All files passed the check.
```

If a file is reported as `BROKEN`, delete it and run the script again.

If you started the download with `nohup`, follow it with:

```
tail download.log                       # latest progress
ls raw_data/*.fastq.gz | wc -l          # should reach 32
```

Sample information (`DEG/pdata.csv`)

`DEG/pdata.csv` is already in the repository. It tells R which sample is wild type and which is mutant. To make it yourself:

1. Open https://www.ebi.ac.uk/ena/browser/view/PRJNA714667.
2. In **Show Column Selection**, tick **run_accession**, **experiment_accession** and **sample_title**.
3. Click **TSV** to download the table. It is tab-separated, so convert it to comma-separated:

```
tr '\t' ',' < ~/Downloads/filereport_read_run_PRJNA714667_tsv.txt >
DEG/pdata.csv
```

It needs a **`run_accession`** column (the SRR IDs) and a **`sample_title`** column containing `ECWT` or `ECMS`.

Step 2 – Reference genome

```
bash scripts/get_reference.sh
```

The reference is *E. coli* K-12 MG1655, assembly ASM584v2 (NCBI page). The script downloads and unzips two files into `reference/`, and skips the download if they are already there:

File	Content
`GCF_000005845.2_ASM584v2_genomic.fna`	Genome sequence (FASTA)
`genomic.gtf`	Gene annotation: where each gene is

Step 3 – Quality control

```
bash scripts/fast_qc.sh
```

FastQC writes a quality report for every FASTQ file into `raw_data/fastqc_results/`. **MultiQC** combines them into one page, `raw_data/fastqc_results/multiqc_report/multiqc_report.html`. Open it in a web browser.

FastQC section	What a good result looks like
Per base sequence quality	Most bases above Phred 30. A drop at the ends of reads means trimming is needed.
Per sequence quality scores	Most reads have a high average quality.
Per base sequence content	Some bias in the first few bases is normal; it should even out after trimming.
Adapter content	No adapters. If adapters are present, they must be removed before alignment.
Per sequence GC content	One peak near the GC content of *E. coli* (about 51 %). Extra peaks can mean contamination.
Sequence duplication levels	Low to moderate. High duplication can mean PCR bias or over-sequencing.

Step 4 – Trimming

```
bash scripts/trim.sh
```

Trim Galore removes adapters and low-quality bases from the ends of the reads. With the settings used here it also:

- keeps the two reads of each pair together (`--paired`);
- discards pairs where a read is shorter than 36 bases after trimming (`--length 36`);
- runs FastQC on the trimmed reads (`--fastqc`), so you can compare before and after.

Samples whose FASTQ files are missing are skipped with a message, so finish Step 1 first.

For each sample, `raw_data/trimmed_reads/` gets:

File	Content
`<sample>_1_val_1.fq.gz`, `<sample>_2_val_2.fq.gz`	Trimmed reads (used in Step 5)
`*_trimming_report.txt`	How many adapters and bases were removed
`*_val_*_fastqc.html`	FastQC report of the trimmed reads

Step 5 – Indexing, alignment and gene counting

```
bash scripts/ecoli_gene_quantification.sh
```

Indexing. An aligner cannot compare every read with the whole genome base by base; that would be far too slow. STAR first builds an **index** of the genome, which lets it find where each read belongs very quickly. The index is built once into `raw_data/star_index/` and reused for every sample; if it already exists, the script skips this step.

Alignment and counting. STAR then aligns each pair of trimmed reads. With `--quantMode GeneCounts` it also uses the GTF to count how many reads fall on each gene, so alignment and counting happen in one step.

Option	Meaning
`--runThreadN 8`	Number of CPU cores (set by `THREADS` at the top of the script).
`--sjdbOverhang 149`	Read length minus 1 (the reads are 150 bases).
`--genomeSAindexNbases 10`	Needed for small genomes like *E. coli*. Without it, STAR can run out of memory or crash.
`--readFilesCommand zcat`	Lets STAR read the compressed `.fq.gz` files directly.
`--outSAMtype BAM SortedByCoordinate`	Writes a sorted BAM file.
`--quantMode GeneCounts`	Writes the gene counts (`ReadsPerGene.out.tab`).

Each sample gets these files in `raw_data/star_align/`:

File	Content
`<sample>_Aligned.sortedByCoord.out.bam`	Aligned reads
`<sample>_ReadsPerGene.out.tab`	Reads counted per gene (used in Step 6)
`<sample>_Log.final.out`	Mapping summary

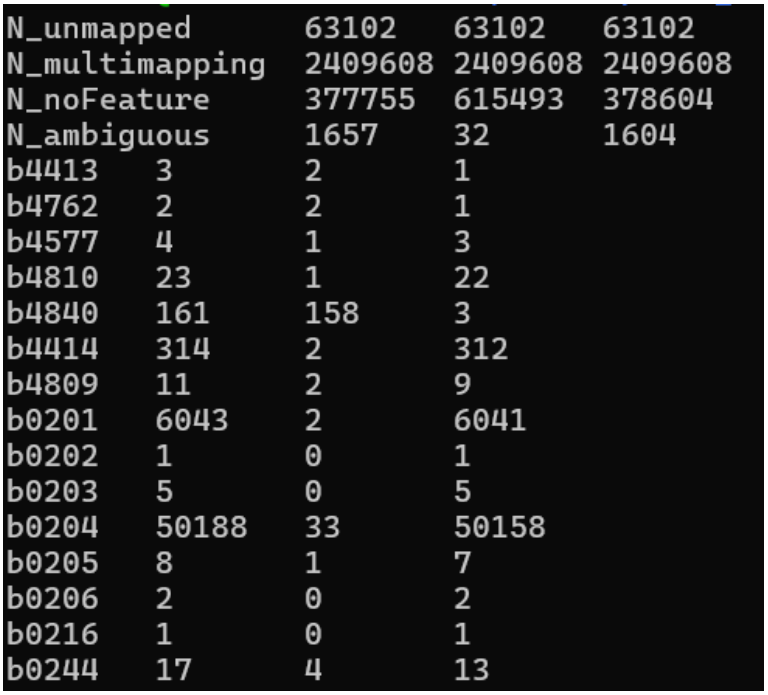
At the end, the script prints the mapping rate of every sample (the `Uniquely mapped reads %` line of each `Log.final.out`). The paper reports about **79–85 %** of reads mapped.

Understanding `ReadsPerGene.out.tab`

Each `ReadsPerGene.out.tab` file has four columns:

Column	Content	Use it when the library is…
1	Gene ID (for example `b0204`)	—
2	Counts ignoring strand	unstranded
3	Counts for reads on the same strand as the gene	forward-stranded
4	Counts for reads on the opposite strand to the gene	reverse-stranded

The first four rows (`N_unmapped`, `N_multimapping`, `N_noFeature`, `N_ambiguous`) are summary numbers, not genes.



```
N_unmapped      63102   63102   63102
N_multimapping  2409608 2409608 2409608
N_noFeature     377755  615493  378604
N_ambiguous     1657    32      1604
b4413   3       2       1
b4762   2       2       1
b4577   4       1       3
b4810   23      1       22
b4840   161     158     3
b4414   314     2       312
b4809   11      2       9
b0201   6043    2       6041
b0202   1       0       1
b0203   5       0       5
b0204   50188   33      50158
b0205   8       1       7
b0206   2       0       2
b0216   1       0       1
b0244   17      4       13
```

The first rows of a ReadsPerGene.out.tab file from this dataset.

Which column to use? Compare columns 3 and 4: if they are similar, the library is unstranded (column 2). If column 3 is much larger, it is forward-stranded (column 3). If column 4 is much larger, it is reverse-stranded (column 4).

> **In this dataset, use column 4.** Column 3 is close to zero and column 4 is almost equal to column 2 (gene `b0204`: 50188, 33, 50158). The library is **reverse-stranded**.

Check a whole file at once:

```
tail -n +5 raw_data/star_align/SRR13970433_ReadsPerGene.out.tab | \
  awk '{c2+=$2; c3+=$3; c4+=$4} END {print "col2:", c2, " col3:", c3, "
col4:", c4}'
```

Step 6 – Merge the gene counts

```
Rscript scripts/merge_data.R
```

STAR writes one count file per sample. `merge_data.R`:

1. reads all `ReadsPerGene.out.tab` files and prints how many it found (should be **16**);
2. drops the 4 summary rows and keeps the count column set by `count_col` (**4** for this dataset);
3. merges the samples into one table, with genes as rows and samples as columns;
4. adds gene names from the GTF (genes without a name keep their ID);
5. saves `DEG/merged_gene_counts.csv` (gene IDs) and `DEG/merged_counts_annotated.csv` (gene names).

Step 7 – Differential expression

```
Rscript scripts/DEG_ecoli.R
```

The script first matches the samples in the count table to `DEG/pdata.csv` and prints how many samples are in each group (should be **8 and 8**). It then removes genes with very few reads (genes are kept only if at least 3 samples have 10 or more reads) and runs **DESeq2**, which tests every gene for a difference between wild type and mutant. DESeq2:

1. **normalises** the counts, so deeply sequenced samples don't look as if they have higher expression;
2. **estimates the variability** of each gene between replicates;
3. **tests** each gene for a difference between the groups;
4. **corrects** the p-values for testing thousands of genes at once. The corrected value is `padj`.

The wild type (ECWT) is the reference group, so the results describe the mutant compared with wild type.

Outputs (in `DEG/`)

File	Content
`DESeq2_results.csv`	All genes, sorted by adjusted p-value
`DESeq2_significant.csv`	Genes with `padj < 0.05` and |log2FC| ≥ 1
`volcano_plot.png`	Volcano plot
`heatmap_top50.png`	Heatmap of the 50 most significant genes

File	Content
`PCA_plot.png`, `PCA_data.csv`	PCA of the samples

The script also prints the number of significant genes, split into up and down in ECMS.

Reading the results table

Column	Meaning
`baseMean`	Average normalised count across all samples
`log2FoldChange`	Change in the mutant compared with wild type. **Positive = higher in ECMS; negative = lower in ECMS.** A value of 1 means a 2-fold change.
`lfcSE`	Standard error of the log2 fold change
`stat`	Test statistic
`pvalue`	Raw p-value
`padj`	Adjusted p-value. **Use this one** to call significance (usually `padj < 0.05`).

Reading the plots

Plot	What it shows	What to look for
PCA	Overall similarity between samples (counts after a variance-stabilising transformation)	The two strains form two separate groups. A sample sitting with the wrong group may be mislabelled or a failed library.
Volcano	Fold change (x) against significance (y) for every gene	Genes far left or right **and** high up have large, significant changes. Coloured genes pass both cut-offs (`padj < 0.05`, |log2FC| ≥ 1).
Heatmap	Expression of the top 50 genes in every sample, scaled per gene	Clear blocks of high and low expression that separate ECWT from ECMS.

Questions

1. How many genes are significantly up and down in the *slyA* mutant?
2. Do the samples separate by strain on PC1 of the PCA plot? How much variance does PC1 explain?
3. In `DEG_ecoli.R`, change the contrast to `c("group", "ECWT", "ECMS")`. What happens to the sign of `log2FoldChange`? Does the number of significant genes change?
4. Rerun Step 6 with `count_col <- 2` (unstranded), then Step 7. How do the results change?
5. Look up the top five genes on EcoCyc. Are any of them known SlyA targets?

Troubleshooting

Problem	Likely cause	Fix
Download stops or a file is reported BROKEN	Network interruption or terminal closed	Delete the broken file, rerun `download_fastq.sh` (inside `screen` or with `nohup`)
`Cannot find .../raw_data`	`raw_data/` missing, or script not inside `training/scripts/`	Run `mkdir -p raw_data` from `training/`; keep the scripts in `scripts/`
STAR: cannot open the FASTA or GTF	Reference missing or still compressed	Run `bash scripts/get_reference.sh`
STAR crashes or runs out of memory while indexing	Index too large for a small genome, or a half-built index	Keep `--genomeSAindexNbases 10`; delete `raw_data/star_index/` and rerun
STAR is very slow or the computer freezes	More threads than CPU cores	Lower `THREADS` in `ecoli_gene_quantification.sh` (check cores with `nproc`)
`ls: cannot access '*_1_val_1.fq.gz'`	Trimming not finished	Rerun Step 4 and check `raw_data/trimmed_reads/`
R: `Count files found: 0`	Wrong `training_dir`, or STAR not finished	Fix the path at the top of the R script; check `raw_data/star_align/`
Very few reads counted per gene	Wrong count column	Use `count_col <- 4` for this dataset
DESeq2: `stopifnot(...) are not all TRUE`	SRR IDs in `pdata.csv` don't match the count columns, or `pdata.csv` is still tab-separated	Compare `colnames(counts)` with `pdata$run_accession`; convert the file as in Step 1
DESeq2: `model matrix is not full rank`	All samples ended up in one group	Check `table(pdata$group)` and the `sample_title` column
`could not find function "EnhancedVolcano"`	Package not installed	`BiocManager::install("EnhancedVolcano")`

Data source

The data used in this training are publicly available:

- NCBI GEO: GSE168963
- ENA / SRA BioProject: PRJNA714667
- Reference genome: *E. coli* K-12 MG1655, GCF_000005845.2 (ASM584v2)

The data set is described in *"Transcriptome RNA Sequencing Data Set of Differential Gene Expression in Escherichia coli BW25113 SlyA Wild Type and Mutant Strains"* (see `Document/`). Please cite the original publication if you use these data.